

HW 7

PHY 611

Fall 2025

Due by end of day on Nov 17

1. a) In scalar QED draw the lowest order in coupling constant 1-loop contribution(s) to the photon propagator. What is the superficial degree of divergence (using a cutoff regulator Λ) of the contribution(s)?

b) Using renormalized perturbation theory with the $\overline{MS}$ scheme, find the value of the counterterms necessary to render the 1-loop photon 2-pt function (i.e. the photon propagator at one loop) finite for scalar QED. To fix notation we'll define the wave function renormalization factors relating the bare and renormalized quantities as:

$$\phi^0 = \sqrt{Z_2}\phi, A_\mu^0 = \sqrt{Z_3}A_\mu, e_0 = Z_e e_R, \text{ and } m_0 = Z_m m_R$$

where $Z_i = 1 + \delta_i$ is the usual expansion for the counterterm. The bare Lagrangian for scalar QED that we'll consider is:

$$-\frac{1}{4}F_{\mu\nu}^2 + |D_\mu\phi_0|^2 - m^2|\phi_0|^2$$

with

$$D_\mu\phi_0 = \partial_\mu + ie_0A_\mu^0\phi_0.$$

You may or may not find the following definitions useful in your calculations

$$Z_1 \equiv Z_e Z_2 \sqrt{Z_3}, \text{ and } Z_4 \equiv Z_e^2 Z_2 Z_3$$

when you expand to define counterterm vertices.

2. **Schwartz problem 11.1** (if you don't have the book I can send you the problem)
3. **Schwartz problem 11.2** (if you don't have the book I can send you the problem)